FINAL PROJECT FOR PHST 620, FALL 2025

(11 Questions and 9 Pages)

Turn in this 1) PDF Format of the completed exam with your name in the header 2) your SAS code and 3) log file, and output file by email to [redacted] and upload them in Blackboard as well.

Due date: By 11.59 PM (EST) on Monday, December 01st. Materials emailed/uploaded after that time will not be accepted.

Here is the scenario: You are a new data analyst recently hired by an outcomes research group within the University of Louisville Hospital Department of Cardiac Surgery. You need to analyze a data set, answer several questions, and provide tables for the principal investigator, and return all documentation so that she can present the material at a professional meeting.

There are two data sets containing data on patients seen by the Cardiothoracic Surgical Team, also called the CST Team, during the study period. Assume that the protocols were approved by the Institutional Review Board (IRB) of the hospital.

DATASETS

SURGERIES DATA SET: There are 9,061 patient records in this data set reflecting 8,092 unique patients.

The variables in this data set are.

- PT_ID (a unique patient ID)
- EVENT_DATE (as SAS date)
- SEX (M or F)
- AGE (age on the EVENT_DATE. Variable AGE is in two decimal years. Some data is missing.)
- RACE (1 and 2 =Native American, 3=African American non-Hispanic, 4=Hispanic, 8=Asian, 9=White non-Hispanic)
- OUTCOME (1=death in hospital, 2 or 9 as discharged)
- CONDX (various surgical procedures)

In this data set, the CST team performs a variety of surgical procedures on patients of all ages.

Each record reflects a single surgical procedure (CONDX) for a patient on any unique day (EVENT_DATE). A patient can have more than one procedure on any given day. However, the data can only record one procedure per line. Therefore, to properly count the number of surgical procedures, you will need to determine which days when combination procedures are performed.

For example, some patients may have had both an MI/BYPASS and an MI/PERC on the same day. Other patients may have had MI/BYPASS and VALVE. In a step in the instructions, you will be asked to determine all combinations.

Another issue is missing values of variable AGE. AGE is recorded in the first patient record on the first event date. AGE must correctly reflect the actual age on the date of the event (EVENT_DATE). You cannot simply retain the AGE in the first record to the following records having different EVENT_DATEs. So, you will need to figure out how to use AGE and EVENT_DATE on the first patient record to calculate AGE on later dates by comparing EVENT_DATEs. RETAIN will help here. Also, to convert days to years, use 365.25 as a factor. **(Hint: Quiz 4).**

SMOKING.sas7bdat data set: The patients were surveyed by the CST Team asking if subjects had any history of smoking tobacco. Any lifetime smoking, regardless of age and current smoking status would be noted if a person smoked at least 100 cigarettes in any 12-month period during his/her lifetime. History of smoking is found on a second data set with the variables, PT_ID, and HOSMKG (1=yes, 0=no).

To answer some of the questions regarding the history of tobacco smoking, you will need a data set that has merged the SURGERIES and SMOKING data sets by PT_ID. Call this data set, COMPLETE.sas7bdat. If you merge the data correctly, all records should correctly reflect the variable per PT_ID without missing data.

So, the following tasks must be done to create a permanent data set called COMPLETE.sas7bdat for analysis that will have 9,061 records with complete data with the following variables (in any order):

- PT_ID (a unique patient ID)
- EVENT_DATE
- SEX (M or F)
- AGE on the EVENT_DATE (Eventually, your data should not have any missing age data, with age values correctly calculated to the EVENT_DATE.)
- RACE (1 and 2 =Native American, 3=African American non-Hispanic, 4=Hispanic, 8=Asian, 9=White non-Hispanic.
- OUTCOME (1=death in hospital, 2 or 9 as discharged)
- CONDX (various surgical procedures)
- HOSMKG (1=Yes, 0=No)

Then, answer several questions, either by filling in the table values or by providing tables generated in ODS RTF that you will cut and paste into the report. I will give you templates on how the correct results (values) should appear.

Questions:

Hint: Before writing the code, please create a permanent library called **PROJECT**.

1) Determine the first event date (MM DD, YYYY) in the study data? Type in the date below. [*3 marks*] 01/01/2003

2) Determine the last event date (MM DD, YYYY) in the study data.? Type in the date below. [*4 marks*] 12/31/2003

3) Count how many missing values are in the variable AGE. [*3 marks*]

   There were 969 values missing from the AGE variable in the original surgeries dataset, which were filled in the complete dataset.

4) Fill in all missing values of AGE as guided in the instruction. [*10 marks*]

   Done. Check the "Cleaning the Data" Section of the code output.

```
PROC SORT DATA=PROJECT.SURGERIES;
        BY PT_ID EVENT_DATE;
RUN;

DATA SURGERIES2;
        SET PROJECT.SURGERIES;
        BY PT_ID EVENT_DATE;
        RETAIN BASE_AGE BASE_DATE;

        IF FIRST.PT_ID THEN
                DO;
                        BASE_AGE=AGE;
                        BASE_DATE=EVENT_DATE;
                END;
        AGE=BASE_AGE +((EVENT_DATE-BASE_DATE)/365.25);
RUN;
```

5) Using PROC FREQ, **complete this table** with demographics on the 8,092 unique patients from data at the first event date record. [*10 marks*]

Table for Q5

Col. 1	Col. 2	Col. 3	Col. 4	Col. 5	Col. 6	Col 7
Race/ethnicity	Number (N)	Race (% of Total)	Male, (n)	Male, (% of Col 2)	Ever smoked, (n)	Ever smoked, (% of Col 2)
Black/African American (non-Hisp)	4,240	52.40	2,430	57.31	1,938	45.71
White non-Hispanic	3149	38.91	2166	68.78	1050	33.34
Hispanic	606	7.49	432	71.29	269	44.39
Asian	82	1.01	56	68.29	13	15.85
Native American	15	.1854	14	93.33	6	40.00
Total	8,092	100	5098	63.00	3276	40.48

6) Write a PROC TABULATE step that recreates this table below with the appropriate values. As a quality check, some of the expected values in the table are shown. Use ODS RTF to output your table, then paste your table into the space below. Make sure that zeroes are present where appropriate so that there are no missing cell values. Also, make sure that you have created formats for RACE and age at first surgery. Hint: You might want to create a new variable, such as AGECAT, to categorize AGE into the age groups, and then apply a format to label the output. [*10 marks*]

Table for Q6

	Age at first surgery							
	under 1	1 -under 5	5 -under 18	18 -under 30	30 -under 50	50 -under 65	65 -under 75	75 and older
	N	N	N	N	N	N	N	N
RACE								
Native American	0	0	0	0	0	11	3	1
Afr Amer (non-Hisp)	22	7	32	304	1902	1464	377	132
Hispanic	0	0	0	6	247	275	62	16
Asian	0	0	0	0	0	5	43	34
White(non-Hisp)	0	0	0	3	529	1508	821	288
All	22	7	32	313	2678	3263	1306	471

Paste your version below:

	Age at first surgery							
	under 1 N	1 -under 5 N	5 -under 18 N	18 -under 30 N	30 -under 50 N	50 -under 65 N	65 -under 75 N	75 and older N
RACE								
Native American	.	.	.	.	.	11	3	1
Afr Amer (non-Hisp)	22	7	32	304	1902	1464	377	132
Hispanic	.	.	.	6	247	275	62	16
Asian	.	.	.	.	.	5	43	34
White(non-Hisp)	.	.	.	3	529	1508	821	288
All	22	7	32	313	2678	3263	1306	471

7) Complete this table with demographics on the 8,092 unique patients from data at the first event date record. Some values are given to you as a guide to your own work. You can recreate the table using PROC TABULATE (easier) or fill in the cells yourself. [*10 marks*]

Table for Q7

		SEX		Ever smoked	
	All	**F**	**M**	**No**	**Yes**
	PctN	**PctN**	**PctN**	**PctN**	**PctN**
AGE at first surgery					
under 1	.27	54.55	45.45	100	0
1 -under 5	.0009	42.86	57.14	100	0
5 -under 18	.40	43.75	56.25	81.25	18.75
18 -under 30	3.87	39.62	60.38	27.80	72.20
30 -under 50	33.09	38.42	61.58	44.81	55.19
50 -under 65	40.32	35.83	64.17	62.89	37.11
65 -under 75	16.14	35.68	34.32	77.03	22.97
75 and older	5.82	37.58	62.42	88.32	11.68

8) **The principal investigator needs to know the maximum number of surgeries that may have occurred on a single day to anyone patient. She wants to see the SAS code and output to confirm her suspicion that any patient had no more than two surgical procedures in any one day.** (Hint: One good tool would be to create a counter.) Label the section in your code for Q8 in a Title line. In the space below, copy/paste a PROC FREQ step and the ODS RTF output that shows the number and frequency of single day and combination day surgeries. For example, the output of a counter would be sufficient. [*15 marks*]

COUNTER	Frequency
The number of single surgeries per day per patient	7791
The number of two surgeries per day per patient	635

SAS CODE for Step 8

```
PROC SORT DATA=PROJECT.COMPLETE;
   BY PT_ID EVENT_DATE;
RUN;

DATA SURGPER_DAY;
   SET PROJECT.COMPLETE;
   BY PT_ID EVENT_DATE;
   RETAIN SURG_CNT;

   IF FIRST.EVENT_DATE THEN SURG_CNT=0;
   SURG_CNT +1;
   IF LAST.EVENT_DATE THEN OUTPUT;
   KEEP PT_ID EVENT_DATE SURG_CNT;
RUN;


PROC FREQ DATA=SURGPER_DAY;
   TABLES SURG_CNT /NOCUM NOPERCENT;
   TITLE "Frequency of Surgeries/Day per Patient";
RUN;
```

SURG_CNT	Frequency
1	7791
2	635

9) Use SAS code and PROC FREQ or TABULATE to fill in this first table of combinations of surgical procedures. (I've provided you with the first column of singleton procedures, those not done in combination, and the Total in the bottom right corner.) You will need to come up with a method that develops the variable CONDX2, reflecting the additional procedure done on the same day. [*15 marks*]

Table of CONDX by CONDX2									
CONDX(CONDX)- First procedure	**CONDX2(CONDX2) –Second procedure**								
Frequency		**ASD**	**Echo only**	**MI/BYPASS**	**MI/PERC**	**ToF**	**VALVE**	**VSD**	**Total**
ASD	25	0	0	1	4	0	1	0	31
CoA	1	0	0	0	0	0	0	0	1
Echo only	30	0	0	0	0	0	0	0	30
Fontan	4	0	0	0	0	0	1	0	5
MI/BYPASS	2894	0	2	0	247	0	35	0	3178
MI/PERC	3972	1	0	238	0	0	37	1	4249
PDA	4	0	0	1	0	1	0	1	7
TGA	8	0	0	0	0	0	0	0	8
ToF	6	0	0	0	1	0	0	0	7
VALVE	827	0	0	25	25	0	10	0	887
VSD	20	0	0	1	2	0	0	0	23
Total	7791	1	2	266	279	1	84	2	8426

Table of CONDX by CONDX2									
CONDX(CONDX)	**CONDX2**								
Frequency Row Pct Col Pct		**ASD**	**Echo only**	**MI/BYPASS**	**MI/PERC**	**ToF**	**VALVE**	**VSD**	**Total**
ASD	25 80.65 0.32	0 0.00 0.00	0 0.00 0.00	1 3.23 0.38	4 12.90 1.43	0 0.00 0.00	1 3.23 1.19	0 0.00 0.00	31
CoA	1 100.00 0.01	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	1
Echo only	30 100.00 0.39	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	30
Fontan	4 80.00 0.05	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	1 20.00 1.19	0 0.00 0.00	5

Table of CONDX by CONDX2									
CONDX(CONDX)	**CONDX2**								
Frequency Row Pct Col Pct		**ASD**	**Echo only**	**MI/BYPASS**	**MI/PERC**	**ToF**	**VALVE**	**VSD**	**Total**
MI/BYPASS	2894 91.06 37.15	0 0.00 0.00	2 0.06 100.00	0 0.00 0.00	247 7.77 88.53	0 0.00 0.00	35 1.10 41.67	0 0.00 0.00	3178
MI/PERC	3972 93.48 50.98	1 0.02 100.00	0 0.00 0.00	238 5.60 89.47	0 0.00 0.00	0 0.00 0.00	37 0.87 44.05	1 0.02 50.00	4249
PDA	4 57.14 0.05	0 0.00 0.00	0 0.00 0.00	1 14.29 0.38	0 0.00 0.00	1 14.29 100.00	0 0.00 0.00	1 14.29 50.00	7
TGA	8 100.00 0.10	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	8
ToF	6 85.71 0.08	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	1 14.29 0.36	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	7
VALVE	827 93.24 10.61	0 0.00 0.00	0 0.00 0.00	25 2.82 9.40	25 2.82 8.96	0 0.00 0.00	10 1.13 11.90	0 0.00 0.00	887
VSD	20 86.96 0.26	0 0.00 0.00	0 0.00 0.00	1 4.35 0.38	2 8.70 0.72	0 0.00 0.00	0 0.00 0.00	0 0.00 0.00	23
Total	7791	1	2	266	279	1	84	2	8426

Then, fill in the table below with the totals for all combination surgeries: [*5 marks*]
(For instance, Total= A & B + B & A):

Combination	Total	Combination	Total
Echo & MI/Bypass	2	Valve & MI/Perc	62
Fontan & Valve	1	PDA & MI/Bypass	1
MI/Bypass & ASD	1	VSD & PDA	1
MI/Perc & ASD	4+1=5	ToF & MI/Perc	1
Valve & ASD	1	ToF & PDA	1
VSD & MI/Perc	3	Valve & Valve (redo)	10
MI/Perc & MI/Bypass	485	VSD & MI/Bypass	1
Valve & MI/Bypass	60		

10) The final step in the instructions is to calculate the mean ages of death and complete the table shown below. Since age is dynamic and changes with each event date, it is important to fill in the missing age with respect to the actual event date on the record.

You may develop your PROC MEANS in ODS RTF and paste your table below. [*10 marks*]

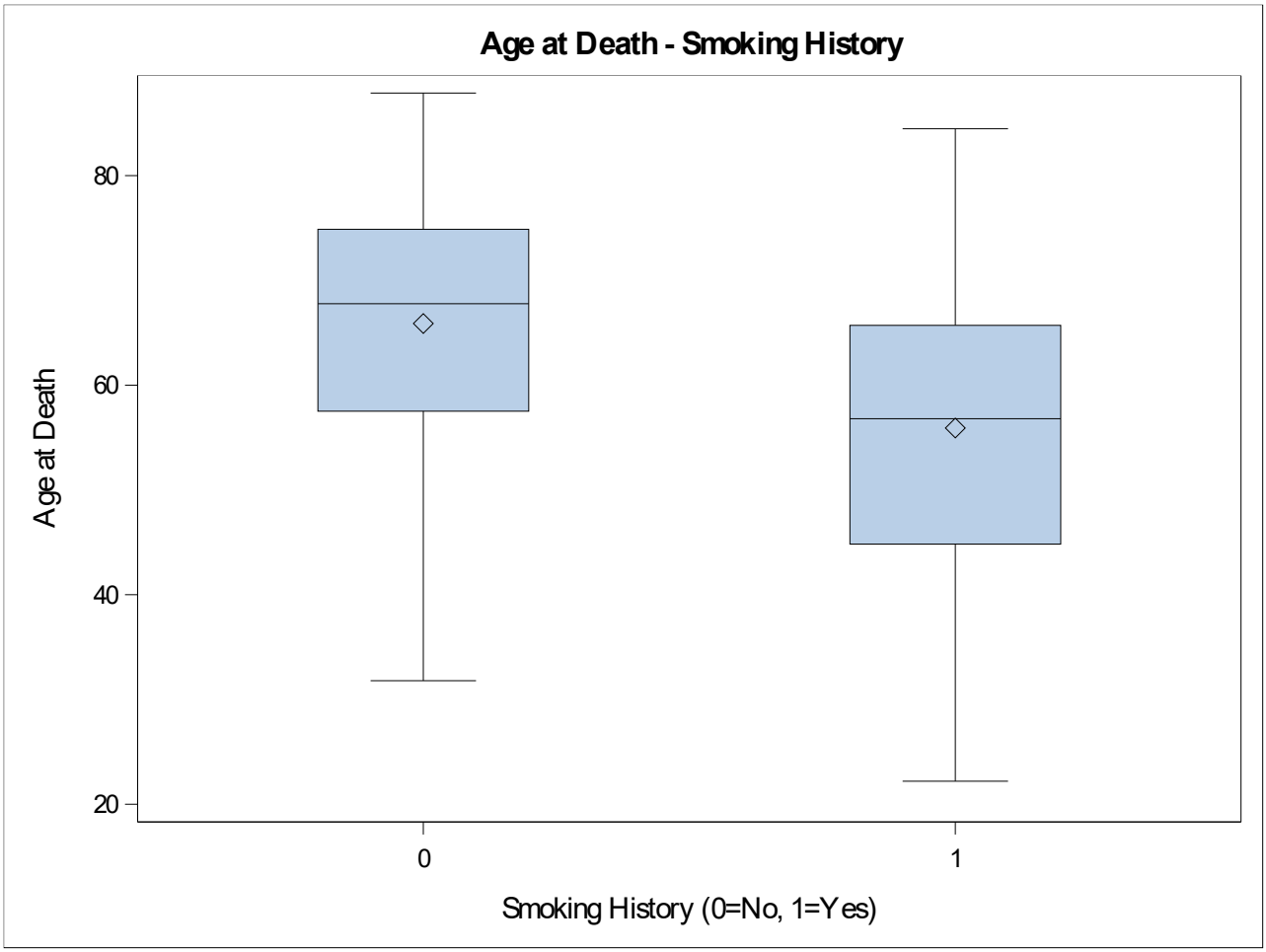
Table Q10: Age at death (age-adjusted for event date compared to age at first event date)

Analysis Variable: AGECARRY Dynamic Age (shown at death)						
Race	**History of smoking**	**N**	**Mean**	**Median**	**Minimum**	**Maximum**
Native American	No	3				
Afr Amer (non-Hisp)	No	10				
	Yes					
Hispanic	No	20				
	Yes					
Asian	No	45				
	Yes					
White(non-Hisp)	No		64.11			
	Yes	49				

Analysis Variable : AGE AGE							
RACE	**HOSMKG**	**N Obs**	**N**	**Mean**	**Median**	**Minimum**	**Maximum**
Native American	No	4	4	72.88	73.26	65.56	79.42
Afr Amer (non-Hisp)	No	10	10	54.36	53.08	31.79	75.82
	Yes	13	13	42.47	43.07	22.20	84.48
Hispanic	No	22	22	56.97	53.43	42.16	87.09
	Yes	23	23	47.78	43.80	34.69	71.14
Asian	No	52	52	73.93	73.93	63.09	87.88
	Yes	10	10	73.54	73.24	62.44	84.05
White(non-Hisp)	No	74	74	64.05	64.94	36.33	86.40
	Yes	53	53	59.40	60.01	38.52	83.78

Why are my counts different? I cannot figure out how to match your table. Hopefully this is enough for partial credit!

11) Draw a box plot of the variable Age for the groups of Smoking and No Smoking on the same axis. Does it appear that smoking influences mortality among these patients? Why? (Hint: Perform two-sample t-test / Week 14_Examples) [*5 marks*]



Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	259	6.25	<.0001
Satterthwaite	Unequal	186.37	6.05	<.0001

Based on this boxplot, and the results of our two-sample t-test, we can conclude that smoking does influence mortality among the patients in our dataset. We arrive at very small p-values, which indicate a statistically significant difference in mortality amongst these groups. Specifically, the mortality is significantly lower in the group that had reported smoking.